



OPEN Environmental variables improve the accuracy of remote sensing estimation of soil organic carbon content

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Accurately and quickly estimating the soil organic carbon (SOC) content is crucial in the monitoring of global carbon. Environmental variables play a significant role in improving the accuracy of the SOC content estimation model. This study focuses on modeling methodologies and environmental variables, which significantly influence the SOC content estimation model. The modeling methods used in this research comprise multiple linear regression (MLR), partial least squares regression (PLSR), random forest, and support vector machines (SVM). The analyzed environmental variables include terrain, climate, soil, and vegetation cover factors. The original spectral reflectance (OSR) of Landsat 5 TM images and the spectral reflectivity after the derivative processing were combined with the above environmental variables to estimate SOC content. The results showed that: (1) The SOC content can be efficiently estimated using the OSR of Landsat 5 TM, however, the derived processing method cannot significantly improve the estimation accuracy. (2) Environmental variables can effectively improve the accuracy of SOC content estimation, with climate and soil factors producing the most significant improvements. (3) Machine learning modeling methods provide better estimation accuracy than MLR and PLSR, especially the SVM model which has the highest accuracy. According to our observations, the best estimation model in the study area was the “OSR + SVM” model ($R^2 = 0.9590$, RMSE = 13.9887, MAE = 10.8075), which considered four environmental factors. This study highlights the significance of environmental variables in monitoring SOC content, offering insights for more precise future SOC assessments. It also provides crucial data support for soil health monitoring and sustainable agricultural development in the study area.

Keywords Soil organic carbon, Landsat 5 TM, Environmental variables, Model comparison, Northeast China transect

Soil is the largest reservoir of organic carbon in terrestrial ecosystems^{1,2}, storing twice the amount of organic carbon in the atmosphere and three times as much as that in vegetation^{3–5}. Therefore, soil organic carbon (SOC) is an important component of the global carbon cycle⁶ and is a fundamental basis for investigating atmospheric CO₂ content and climate change. Moreover, SOC can effectively improve soil fertility by sequestering carbon and participating in soil nutrient cycling^{7,8}. It improved soil nutrient and water retention capacity, alongside promoting soil respiration, thereby slowing down land degradation and ensuring food production security^{9–11}. As a result, accurate and rapid monitoring of SOC content is essential for global carbon pool monitoring, climate change research, and national agricultural management policy formulation.

Traditional methods for estimating SOC content rely on field sampling and laboratory physicochemical analyses. The results of these methods are accurate and reliable, although the process of collecting sampling points is time-consuming and labor-intensive. Besides, it can cause some damage to the surface of the area where the points were collected^{12,13}. Furthermore, it is economically expensive and difficult to monitor the spatial distribution of SOC over a large region¹⁴. Therefore, the key issue to be addressed is how to estimate the SOC content of larger regions in a non-destructive, rapid, and accurate. Recently, the development of remote sensing has provided new methods for the non-destructive monitoring of SOC content in a larger region. Many scholars have constructed simple linear estimation models for the SOC content using spectral bands or their combinations based

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on the spectral response characteristics of soils with different SOC contents.^{13,15,16} For example, Zhou et al.¹⁷ compared the differences in SOC response between different optical and radar sensors and used their response characteristics to predict SOC content in Spain based on RF. Allory et al.¹⁵ discussed the differences between soil spectral profiles of urban and rural areas and predicted SOC stocks for two cities in France. However, these methods mostly use the original spectral reflectance of each band and their combinations as variables, and their prediction accuracy is not high despite their simple calculation process and strong interpretability. Therefore, it is important to research models that incorporate environmental factors (e.g., climate, terrain, soil, and vegetation cover) to improve the accuracy of SOC content estimation models based on optical remote sensing.

Many studies have demonstrated that SOC exhibits significant spatial heterogeneity^{18,19} and its content is strongly influenced by environmental variables, including climate, terrain, and soil, particularly precipitation^{8,20–23}. For example, Luo et al.²⁴ constructed a prediction model for the soil organic matter content in the Songnen Plain of China based on random forest and found that the model prediction accuracy was significantly affected by precipitation. Meanwhile, Luo et al.²⁵ examined the best variable for forecasting the soil organic matter during the bare soil period in Northeast China and concluded that climatic factors can effectively improve the predictive accuracy. These studies showed that incorporating environmental variables as auxiliary variables to the SOC content estimation model, especially those that have a more significant impact on SOC changes, can improve the accurate representation of the spatial distribution of SOC, and it is an important approach to enhance the current estimation model. However, adding more variables can easily lead to problems such as multicollinearity, the number of variables is larger than the samples, and it requires a high level of predictive model arithmetic power²⁶. Among the modeling methods for estimating SOC content, PLSR resolves the problem of variable multicollinearity and works with a smaller sample size than MLR, while machine learning regression models like RF and SVM provide great computational ability for complex models^{27–31}. At the same time, most of the current studies primarily concentrate on the impact of individual environmental factors or modeling methods on the estimation accuracy of SOC content, and there is a lack of comparing the effects of various environmental variables on the estimation accuracy using different methods. Therefore, it is crucial to ascertain the optimal environmental variables and modeling methods for models estimating the SOC content.

The terrestrial transect has already become an important approach for the International Geosphere-Biosphere Programme (IGBP) to study global change. Its aim is to understand the response of terrestrial ecosystems to global change, and to predict and evaluate the potential impacts of global change on terrestrial ecosystems³². The Northeast China Transect (NECT) was one of the first IGBP set up for studying global change study in 1993. It is a gradient affected mainly by precipitation³². The NECT follows an evident precipitation gradient from east to west, creating various vegetation types and soil characteristics that reflect the most significant and critical climate changes in the mid-latitude temperate zone of East Asia. Therefore, non-destructive monitoring of SOC content in NECT can effectively demonstrate the changes to environmental variables that affect SOC content. This is crucial for the rapid and effective estimation of SOC content over a large region.

Derivative processing is an effective method for eliminating information interference, yet it is rarely applied to Landsat 5 image. In this study, we focused on using Landsat 5 images processed in various ways as the primary data, with environmental variables serving as auxiliary factors, to develop a model for rapidly estimating soil organic carbon content in the Northeast Transect in 2001. We hypothesized that Landsat 5 images could indirectly estimate SOC content, and that derivative processing could effectively remove background interference. However, we also posited that second-order derivation might inhibit the expression of information regarding SOC content, potentially reducing the accuracy of the model's estimation. The specific objectives of this study are: (1) To compare the effects of different environmental variables on the accuracy of the SOC estimation model. (2) To evaluate the SOC estimation capability of Landsat 5 TM images before and after derivative processing. (3) To analyze the differences in optimal variables for constructing SOC estimation models using MLP, PLSR, RF, and SVM methods. (4) To identify the best variables and methods for SOC estimation and map the spatial distribution of SOC content in the study area.

Materials and methods

Study area

The Northeast China Transect (NECT) (112°–130° 30' E), with a total length of about 1600 km, is located in the mid-latitude semiarid region along 43° 30' N (Fig. 1) and serves as a land sample zone for global change research wherein precipitation is the main driver. The NECT represents a gradual transition from an oceanic humid climate to a continental arid climate, extending from east to west, and characterized by three climatic zones, namely temperate humid zone, temperate humid semiarid zone, and temperate semiarid and arid zone. Meanwhile, the moisture gradient in the study area is extremely decreased from east to west. In 2001, the annual precipitation in the Changbai Mountains in the east was as high as 800 mm, while that in the central agricultural area was about 550 mm, that in the steppe zone in the center-west ranged from 350 to 500 mm, and that in the western desert steppe was less than 200 mm (<https://data.cma.cn/>). The average annual temperature ranged from −3.86 to 7.28 °C in the region (<https://data.cma.cn/>), and the differences in heat across the region are mainly influenced by the topography with the central plains having the highest average annual temperature and the mountainous or plateau areas at the east and west ends being cooler. Besides, the land use pattern and intensity in the NECT exhibit distinct spatial variations, and there is a complete sequence and transition of forest-agriculture-pasture areas from east to west.

Collection and laboratory determination of soil samples

The field trip and soil samples collection took place from July 26 to August 8, 2001. The route began from the Chunhua Forest Farm in Hunchun, Jilin, China, and proceeded westward along 43° 30' N. During this route, a

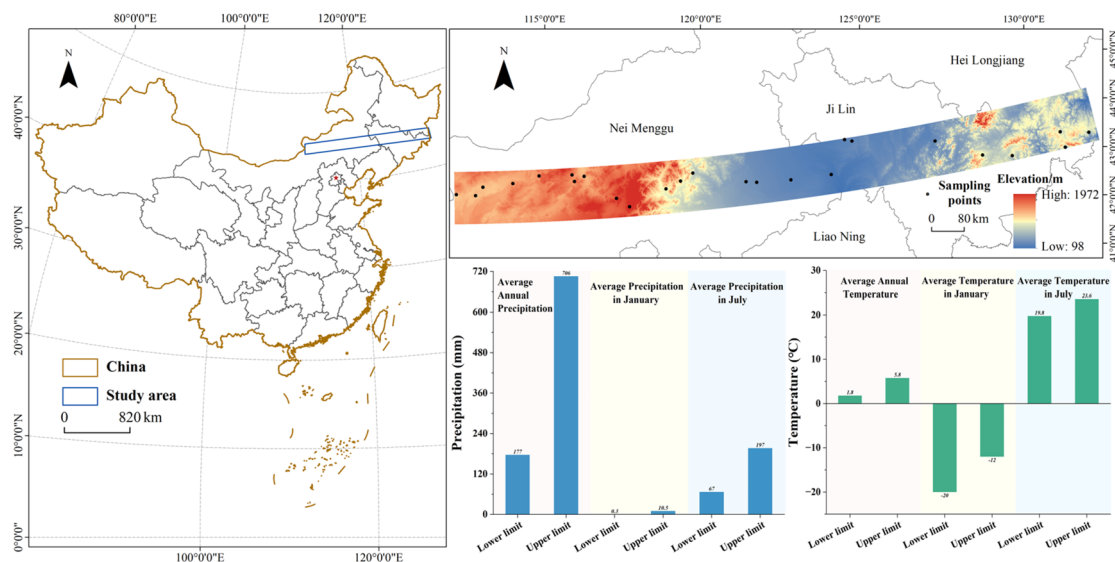


Figure 1. Climatic and geographic of the study area and spatial distribution of sampling points.

total of 25 soil samples were collected. Their distribution of sampling points is shown in Fig. 1. In the process of sampling, one soil sample was collected using the five-point sampling method at a depth of 0–20 cm within 30 × 30 m. Each soil sample weighed about 1 kg. At the same time, to locate the sampling points and determine their corresponding positions in Landsat 5 TM images, the coordinates of the sampling points were established using the Magellan GPS Field PROVTM (California, U.S.A.), which is a Global Positioning System (GPS).

In addition, during the collection of soil samples, we removed coarse stones, plant fragments, and roots, placed them in self-sealing bags, and brought them back to the laboratory for analysis. The soil samples were air-dried, ground, filtered (2 mm), and analyzed indoors. Afterward, the SOC content was determined using external heating and the potassium dichromate volumetric method³³, while soil pH was determined using a pH meter with a soil–water ratio of 1:5.

Landsat 5 TM image acquiring and preprocessing

28 Landsat 5 TM images covering the study area were collected and screened from the Geospatial Data Cloud website (<http://www.gscloud.cn/>) based on the sampling time and cloud conditions of the soil sample sites in this study. The images were pre-processed using ENVI (The Environment for Visualizing Images, Version 5.3, <http://www.harrisgeospatial.com/>) software³⁴, specifically radiometric calibration, atmospheric correction, image stitching, and clipping. During the radiometric calibration process, the thermal infrared bands were not included in the calculations due to their different resolutions. Following radiometric calibration, only six bands of images remain (bands 1–5 and 7), all of which have a spatial resolution of 30 m³⁵. Moreover, the preprocessed original spectral reflectance was subjected to first-order derivation (FD) and second-order derivation (SD) through ENVI 5.3 software.

Data sources and processes for environmental variables

Surface reflectance results from the interaction of various surface factors¹³. The SOC content estimation model based on spectral reflectance lacks the influence of other surface factors. Consequently, its prediction fails to reflect actual scenarios, leading to significant prediction errors. In order to make the estimation results better match actual conditions, this study considered multiple environmental factors, such as climatic conditions, topographic differences, soil physicochemical properties, and vegetation cover on soil spectral reflectance.

Climate factors (CF)

Average annual precipitation and average annual temperature were chosen to characterise the climate of the study area. The two variables have a crucial contribution to the input and decomposition of SOC, and they significantly affect SOC storage^{25,31}. The climate factors in this study were obtained from the National Earth System Science Data Centre (<http://www.geodata.cn/>), and its spatial distribution in the study area was mapped by ArcGIS (Geographic Information System, Version 10.2, <http://www.esri.com/software/arcgis>) software³⁶, with a sample resolution of up to 30 m using the resampling method (Fig. 2a,b).

Terrain factors (TF)

Terrain factors have a significant impact on hydrological and ecological mechanisms, like surface runoff, plant growth, and distribution, which in turn directly impacts the SOC content^{37,38}. Among them, elevation, slope, and slope orientation are frequently utilized to indicate the topographical condition of a region. Therefore, these factors were selected to accurately represent the topographic features of the study area. The terrain factors

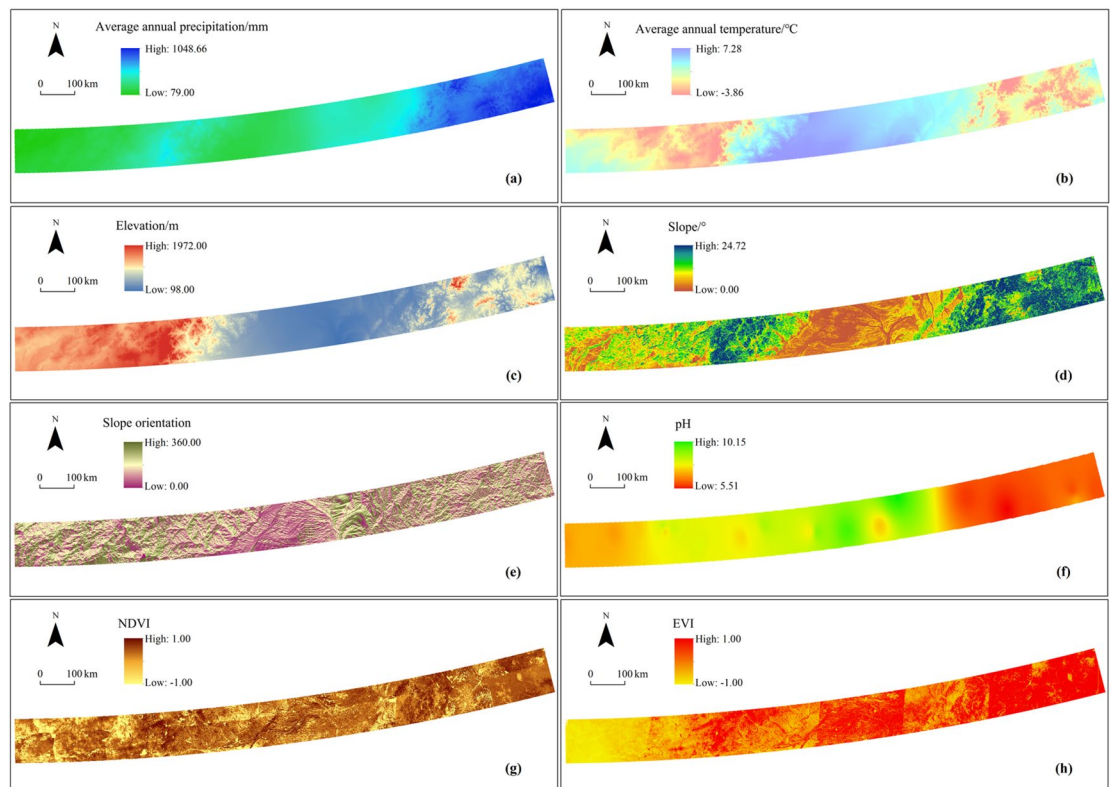


Figure 2. The environmental variables selected for this study, 2001. [(a) is the spatial distribution of average annual precipitation. (b) shows average annual temperature. (c) shows the spatial distribution of elevation. (d) shows the spatial distribution of slope. (e) is the spatial distribution of slope orientation. (f) shows soil pH. (g) is the spatial distribution of NDVI. While (h) shows EVI].

of the study area were obtained by using the spatial analysis tool of ENVI 5.3 software to extract data from ASTERGDEM (<http://earthexplorer.usgs.gov/>), and the results are shown in Fig. 2c,e.

Soil factor (SF)

The SOC content is affected at the local scale by soil physicochemical properties³⁷. Soil pH is one of these properties and influences the rate of the SOC decomposition process by shaping the behavior of soil microorganisms, thus modifying its speed³⁹. In acidic soils, microbial species and numbers are constrained, leading to a slower SOC decomposition rate, particularly when the pH is out of the optimal range, that is, below 5.5 or above 8.5, since these values are highly detrimental to microbial growth and activity^{40,41}. The spatial distribution of soil pH in the study area was mapped using the Inverse Distance Weight interpolation (IDW) method of ArcGIS 10.2 software that was based on the sampling points. The spatial resolution was sampled up to 30 m using the resampling method. The processed spatial distribution of soil pH is shown in Fig. 2f.

Vegetation cover factor (VF)

The Normalised Difference Vegetation Index (NDVI) is the most commonly used vegetation index to reflect vegetation cover. The concept of ratio calculation can significantly reduce noise, which effectively represents the plant growth status and the degree of vegetation cover⁴². The greater the plant cover, the more abundant the apoplastic material and leaves, while the residual roots and leaves remain in the soil to be decomposed by microorganisms, thereby increasing the SOC content⁴³. The Enhanced Vegetation Index (EVI) is an improvement of NDVI by adding blue light to correct for atmospheric aerosol scattering and soil background. It reduces the effects of background noise and stably characterises the vegetation condition in high biomass areas⁴⁴. We calculated the NDVI and EVI of the study area using ENVI 5.3 software, based on the preprocessed Landsat 5 TM images (Fig. 2g,h). The specific calculation formulas are described below.

$$\text{NDVI} = (\text{Band4} - \text{Band3}) / (\text{Band4} + \text{Band3}) \quad (1)$$

$$\text{EVI} = 2.5 \times (\text{Band4} - \text{Band3}) / (\text{Band4} + 6 \times \text{Band3} - 7.5 \times \text{Band1}) \quad (2)$$

where Band1, Band3, and Band4 represent the original reflectance of the corresponding bands in the preprocessed Landsat 5 TM images, respectively.

Modeling methods

The multiple linear regression model is widely used in remote sensing inversion studies. In this study, we constructed the SOC content estimation model by employing MLR and PLSR models. Additionally, to address the saturation issue in non-linear fitting, we employed random forest and support vector machines models. The details are presented below.

Multiple linear regression (MLR): MLR is a traditional method for exploring the relationship between dependent and independent variables while effectively determining the influence of multiple independent variables on the dependent variable. In particular, by entering distinct arrangements of independent variables, examining and discerning the deviation between the anticipated and actual values of the dependent variables, and choosing the combination of independent variables with the least prognostication error, the most outstanding combination of regression independent variables under the approach is determined. In this study, the MLR modeling process was performed based on the “stats” package of R 4.2.3⁴⁵.

Partial least squares regression (PLSR): PLSR is a commonly used multivariate linear regression modeling method that analyzes the interdependence of predictor and response variables⁴⁶. PLSR concentrates on principal component analysis, linear regression analysis, and typical correlation analysis, and is suitable for studies with multiple sets of predictor variables and small sample sizes. In this study, the parameter selection and modeling process of PLSR were performed based on the “pls” package of R 4.2.3^{45,47}.

Random forest (RF): RF is a machine learning model in which the set of predictor variables is randomly constrained in each split point, which effectively avoids the problem of tree-to-tree correlation in the computation, and thus achieves highly accurate predictions^{41,48}. Compared with traditional decision tree construction methods, RF can effectively deal with the problem of insensitivity to the multicollinearity of predictor variables. Furthermore, the ease of parameter tuning is another important advantage of RF, in which the number of spanning trees (ntree), the minimum number of leaves (nodesize), and the number of predictor variables used to split the nodes at each node (mtry) are the key parameters to improve the accuracy of the model. In this study, the estimated model ntree was taken as 500, nodesize was set to 5, and mtry was 3. The parameter selection and modeling process were carried out using the “randomForest” package of R 4.2.3^{45,49}.

Support Vector Machines (SVM): In the early days, SVM analysis was mainly applied to classification problems. The algorithm is flexible and efficient by inputting data from multiple variables and creating multiple planes to classify the variables into different categories⁵⁰. Recently, many scholars demonstrated that SVM exhibits good modeling stability and high prediction accuracy^{28,51,52}. Therefore, this study constructed the SOC estimation model based on the “e1071” package of R 4.2.3^{45,53}.

Model accuracy evaluation methods

To assess the accuracy of the model, we used the model decision coefficients (R^2), root mean square error (RMSE), and mean absolute error (MAE). The formulas for R^2 , RMSE, and MAE are described in Eqs. (3–5). Among these, the range of R^2 is between 0 and 1, and a higher value of R^2 indicates a stronger correlation between the measured and the predicted values. In contrast, RMSE and MAE are both in the range of $[0, +\infty]$. The lower the values of RMSE and MAE, the closer the measured values are to the predicted values. On the other hand, larger values of RMSE and MAE indicate higher prediction errors. In this study, the three indicators were calculated based on the “modelr” package of R 4.2.3^{45,54}.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - y'_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (3)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \times \sum_{i=1}^n (y_i - y'_i)^2} \quad (4)$$

$$\text{MAE} = \frac{1}{n} \times \sum_{i=1}^n |y_i - y'_i| \quad (5)$$

where y_i and y'_i represent the measured and predicted SOC content, respectively, $\bar{y}$ is the average of the measured SOC content, and n is the number of samples.

Results

Descriptive statistics of SOC content and environmental variables at sample sites

SOC primarily originates from animal, plant, and microbial residues and their secretions, and undergoes a constant process of decomposition and formation. The SOC content represents the current dynamic equilibrium state of the ecosystem to a certain extent^{55–57}. According to Table 2, the overall SOC content was high in the study area, attaining a mean value of about 22.90 g kg⁻¹. However, the spatial variation of the SOC content was considerably different based on various environmental factors such as climate, vegetation, and soil, ranging from 2.06 to 107.04 g kg⁻¹. The standard deviation of 27.04 g kg⁻¹ and the coefficient of variation (CV) was 118.10%. The regions towards the east of the study area were identified by dense vegetation, high accumulation of apomixis, and gradual breakdown of SOC, leading to an increase in SOC content. Meanwhile, the regions towards the west had reduced SOC content, which could be due to the hastened rate of SOC decomposition caused by the climatic circumstances (Fig. 3). In addition, 25 samples were randomly divided into a training dataset and a validation dataset in this study, with the former including 20 samples and the latter including 5 samples. The

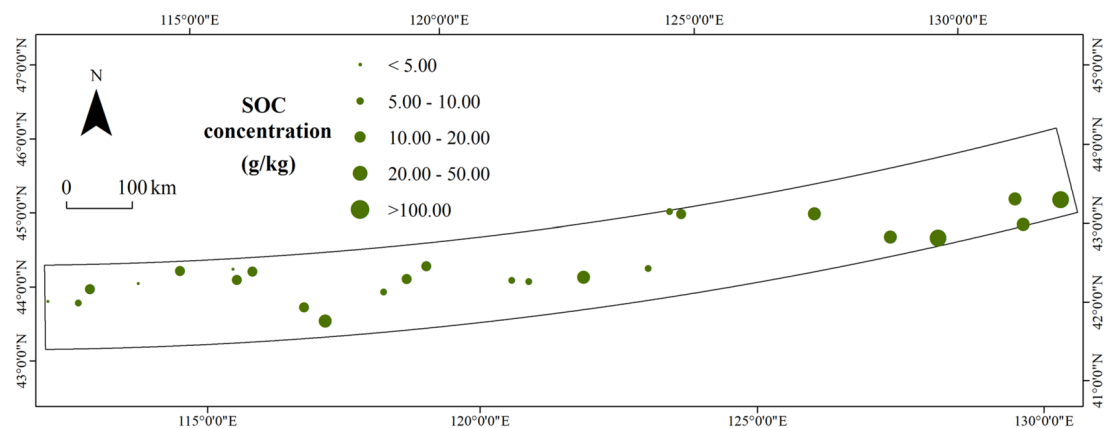


Figure 3. Distribution of SOC content in soil sample sites collected in the study area.

statistical characteristics of both datasets were similar (Table 2). It was therefore feasible to apply the estimation model in this study.

The spatial distribution of SOC content was impacted by environmental variables, however, the effects varied among different classes of environmental variables. Table 1 shows that the terrain in the study area significantly varies. Specifically, the sampling points had an elevation range of 120.00–1580.00 m, with a mean of 722.56 m and a standard deviation of 410.70 m. The slope ranged from 1.01 to 43.39°, with a CV of 104.76%, while the standard deviation of the slope orientation was 107.16%. Furthermore, the NDVI ranged from 0.08 to 0.88, with a mean value of 0.46 and CV of 59.31%, while EVI ranged from 0.13 to 1.00 with a mean value of 0.67 and CV of 47.54%. These observations indicate that the vegetation cover in the study area was high and unevenly distributed. Climatic resources varied significantly, especially the variation of average annual precipitation, ranging from 101.04 to 906.59 mm, with an average of 388.48 mm and CV of 67.34%. The average temperature was about 3.67 °C, with a CV of 51.60%.

Association analysis of spectral reflectance with SOC content

Defining the spectral reflectance characteristics of SOC is an important basis for estimating SOC content. Figure 4 shows the spectral reflectance characteristics of soils at five different SOC levels. In OSR, the spectral characteristics of different SOC contents showed significant variance, with spectral differences mainly occurring in Band1-3, 5, and 7, especially in the lower SOC levels. In the lower level range, the increase in SOC contents led to a significant decrease in reflection across the five bands, consistent with previous research^{16,58,59}. With higher SOC content, the spectral curve differences primarily occurred in Band4, with fewer variations in the other bands. In FD, the spectral differences of lower SOC contents mainly occurred in Band4, Band5, and Band7, while the differences of spectral curves in the bands of higher SOC contents were small. In addition, the differences in different SOC contents are mainly observed in Band7 in SD.

Furthermore, association analysis was used to investigate the relationship between SOC content and spectral reflectance. As shown in Table 2, except for Band1, which was processed by the first-order derivation and had a weak correlation with SOC content, all the other bands had high correlations. Specifically, Band1-3, Band5, and Band7 showed a significant negative correlation with SOC content in OSR, especially Band7 with a correlation coefficient as high as −0.523, while Band4 was significantly positively correlated with SOC content (0.409). After

	Mean	Minimum	Maximum	Standard deviation	CV (%)
Dataset (g kg ⁻¹)	22.90	2.06	107.04	27.04	118.10
Training dataset (g kg ⁻¹)	24.61	4.91	107.04	28.80	117.03
Validation dataset (g kg ⁻¹)	16.09	2.06	48.29	19.41	120.67
Elevation (m)	722.56	120.00	1580.00	410.70	56.84
Slpoe (°)	1.74	0.07	6.03	1.83	104.76
Slope orientation	157.80	3.50	340.98	107.16	67.91
Average annual temperature (°C)	3.67	− 0.34	6.83	1.9	51.60
Average annual precipitation (mm)	388.48	101.04	906.59	261.62	67.34
Soil pH	7.70	5.51	10.15	1.26	16.38
NDVI	0.46	0.08	0.88	0.27	59.31
EVI	0.67	0.13	1.00	0.32	47.54

Table 1. Descriptive statistics between SOC content and environmental elements in the study area.

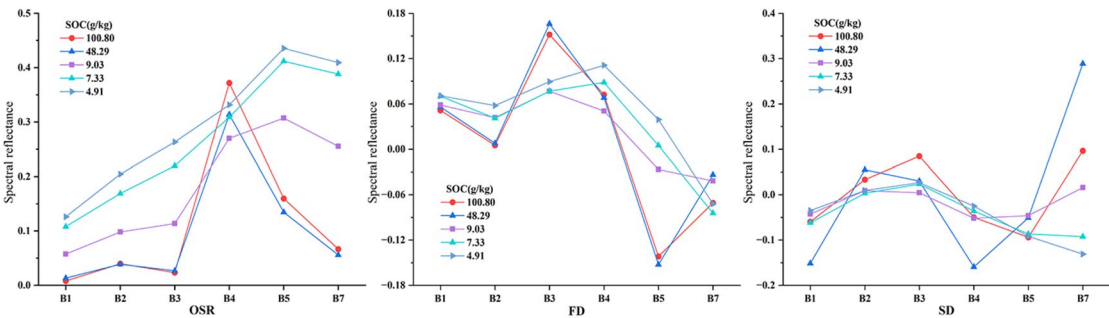


Figure 4. Spectral characteristics of Landsat 5 TM with different SOC contents.

Derivative processing	Band 1	Band 2	Band 3	Band 4	Band 5	Band 7
OSR	−0.467*	−0.430*	−0.478*	0.409*	−0.479*	−0.523**
FD	−0.029	−0.474*	0.594**	−0.272	−0.625***	0.304
SD	−0.429*	0.388	0.474*	−0.371	0.170	0.530**

Table 2. Association analysis between reflectance and SOC content under different derivative treatments. The symbols *, **, and *** represent *p* values of less than 0.05, 0.01, and 0.001, respectively.

the first-order derivation transformation, a significant increase in the correlation between the SOC content and Band2, Band3, and Band5 was noted. Specifically, Band3 depicted a significant positive correlation (0.594), whereas Band5 showed a highly significant negative correlation (−0.625). After the second-order derivation transformation, a significant decrease was observed in the correlation of Band5 with SOC content. Moreover, the correlation between SOC content and the remaining bands was comparatively weaker, especially Band5. In summary, although there was a high correlation between reflectance and SOC content in the original bands, the first-order derivation improved this correlation to some extent. However, the second-order derivation did not demonstrate this ability, which may be related to the lack of information caused by the amplified noise effect after the second-order derivation.

Comparative results of estimation models

Derivative processing

The training and validation results of the 72 estimation models for SOC content are shown in Table 3. The results showed that derivative processing did not significantly improve the estimation accuracy. Moreover, the second-order derivative led to a significant decrease in the estimation ability of the model. In particular, the derivative processing reduced the estimation accuracy for the model constructed using a training dataset. Taking the model without environmental variables as an example, the accuracy of “FD + MLR” decreased by 0.1395 compared to “OSR + MLR”. Also, the accuracy of “SD + MLR” decreased by 0.1708 compared to “OSR + MLR”. Similarly, machine learning methods exhibited the same trend, with “FD + Band + SVM” decreasing the accuracy by 0.0069 compared to “OSR + Band + SVM”, and “SD + Band + SVM” decreasing the estimation accuracy by 0.0828 compared to “OSR + Band + SVM”. Moreover, environmental variables alleviated the negative influence of derivative processing to some extent, while this phenomenon still existed. Consider the model with four environmental variables, where the accuracy of “FD + EV + SVM” decreased by 0.0328 compared to “OSR + EV + SVM”, while “SD + EV + SVM” was 0.0112 less accurate than “OSR + EV + SVM”. It is worth mentioning that terrain factors and derivative processing improved the estimation accuracy of the MLR method. Regarding the “MLR + TF” context, derivative processing improved by 0.0204 (FD) and 0.0882 (SD) respectively.

Environmental variables

The modeling accuracy was significantly higher for the model which takes environmental variables into account in comparison to the spectral model that only considers spectral bands. The addition of VF, CF, TF, and SF individually can improve the accuracy of the model, especially of CF and SF (Table 3). Specifically, the estimation models constructed using the training dataset exhibited better simulation accuracies under various derivative treatments when they incorporated CF, TF, and SF independently. For example, under different derivative processing, the SVM method improved the model with only CF added by 0.1499 (OSR), 0.0839 (FD), and 0.1854 (SD) respectively, when compared to the model with no environmental variables added. In contrast, the accuracy of the model with only TF added improved by 0.1451 (OSR), 0.1305 (FD), and 0.1829 (SD) respectively. By adding only SF, the model accuracy improved by 0.1504 (OSR), 0.0846 (FD), and 0.1888 (SD) respectively. Furthermore, although adding only VF improved the model accuracy, it had less influence compared to the three other types of variables. For example, in OSR, under different modeling methods, adding only VF improved the model accuracy by 0.1202 (PLSR), 0.0211 (RF), and 0.0054 (SVM) compared to the model with no environmental variables, respectively. In addition, there was no significant difference in the accuracy precision of the MLSR method. It is

Derivative processing	Variables	Method	Training dataset			Validation dataset		
			R ²	RMSE	MAE	R ²	RMSE	MAE
OSR	Band	MLR	0.6411	17.2695	13.6362	0.7705	15.6793	13.8521
		PLSR	0.6220	17.2555	14.0568	0.7114	28.7290	22.9947
		RF	0.8708	12.3316	7.7955	0.6935	24.8760	20.6730
		SVM	0.7888	17.1664	9.2434	0.8185	16.4774	13.7986
	Band + VF	MLR	0.7157	15.8626	12.7894	0.8503	17.5478	15.6669
		PLSR	0.5018	19.8100	15.4648	0.8660	22.3776	20.3683
		RF	0.8497	13.2807	8.3085	0.7931	20.7094	17.7127
		SVM	0.7834	17.8919	9.2396	0.8191	14.6860	12.1439
	Band + CF	MLR	0.7678	13.9689	11.0171	0.7545	15.2336	12.6855
		PLSR	0.3753	22.1837	14.5503	0.7874	9.8450	5.9321
		RF	0.8653	12.1718	7.5320	0.7808	20.2388	16.5484
		SVM	0.9387	13.4141	6.4943	0.9129	12.0491	10.8142
	Band + TF	MLR	0.7020	15.5860	12.7408	0.6653	14.8755	11.8535
		PLSR	0.0097	27.9304	18.4565	0.0812	18.2098	16.7097
		RF	0.8843	12.2373	7.4802	0.6783	21.3246	17.8802
		SVM	0.9340	14.4445	6.9974	0.6567	12.6137	12.0627
	Band + SF	MLR	0.7975	13.1935	11.5978	0.7775	15.8934	14.6924
		PLSR	0.7833	13.0660	10.7304	0.6835	30.1850	25.6617
		RF	0.8521	12.3426	7.9868	0.7084	18.0483	14.4850
		SVM	0.9392	13.0253	6.7051	0.9276	9.1782	7.8489
	Band + EV	MLR	0.8540	11.0773	9.2049	0.8541	12.6296	10.1763
		PLSR	0.4059	21.6330	15.0098	0.6302	13.3522	10.0488
		RF	0.8734	12.3335	7.7030	0.8567	14.7745	13.0316
		SVM	0.9590	13.9887	6.8237	0.9220	11.6165	10.8075
FD	Band	MLR	0.5015	19.8452	15.0729	0.5907	12.2626	10.4900
		PLSR	0.3271	23.0229	15.5979	0.4149	15.1717	12.2324
		RF	0.7976	14.4133	9.2378	0.9714	8.6608	7.2537
		SVM	0.7819	19.4025	9.0288	0.0137	17.3158	14.0776
	Band + VF	MLR	0.6702	16.3734	12.2534	0.9071	11.6585	11.2230
		PLSR	0.5094	19.6595	14.8243	0.9413	14.4585	12.1319
		RF	0.7897	14.7824	9.2453	0.9825	8.3018	7.9267
		SVM	0.8294	19.1075	8.2962	0.8086	13.5935	11.3517
	Band + CF	MLR	0.7141	15.1354	11.3645	0.9774	6.8182	5.5528
		PLSR	0.3753	22.1837	14.5503	0.7874	9.8450	5.9321
		RF	0.8191	13.6360	8.3728	0.9318	8.4158	7.8118
		SVM	0.8658	17.1419	7.6876	0.8440	11.6458	10.6011
	Band + TF	MLR	0.7224	14.8210	11.8828	0.6848	10.8595	10.2246
		PLSR	0.0097	27.9304	18.4565	0.0812	18.2098	16.7097
		RF	0.8104	14.1363	8.8841	0.8706	10.9840	10.6474
		SVM	0.9124	16.7893	7.7036	0.0016	17.5374	14.5974
	Band + SF	MLR	0.5840	18.1495	14.1881	0.5975	12.3928	9.9326
		PLSR	0.4904	20.0347	15.2645	0.3538	17.2831	15.5209
		RF	0.7997	13.8580	8.7846	0.9455	7.4883	7.2009
		SVM	0.8665	17.0918	7.3529	0.7851	14.4511	12.8469
	Band + EV	MLR	0.8858	9.8040	7.7685	0.9698	9.5297	8.6771
		PLSR	0.4059	21.6330	15.0098	0.6302	13.3522	10.0488
		RF	0.8285	13.5195	8.5998	0.9963	7.4787	6.6840
		SVM	0.9262	16.2342	7.4829	0.8142	15.3868	13.1409
Continued								

Derivative processing	Variables	Method	Training dataset			Validation dataset		
			R ²	RMSE	MAE	R ²	RMSE	MAE
SD	Band	MLR	0.4703	20.4851	14.9385	0.5339	13.6430	10.6894
		PLSR	0.3461	22.6960	15.9068	0.1732	16.7963	14.4606
		RF	0.7819	13.7332	8.8835	0.6186	14.7054	13.1131
		SVM	0.7060	21.3335	10.3507	0.7488	13.4908	11.8947
	Band + VF	MLR	0.6360	17.0055	13.6279	0.8822	8.9255	8.2698
		PLSR	0.5337	19.1654	14.6666	0.9667	9.2567	8.2606
		RF	0.7864	13.7185	8.8698	0.9633	10.5656	9.6541
		SVM	0.7580	20.4704	9.0312	0.8005	13.8406	11.3059
	Band + CF	MLR	0.6620	16.3954	12.7593	0.8626	8.2269	7.7830
		PLSR	0.3753	22.1837	14.5503	0.7874	9.8450	5.9321
		RF	0.8161	12.9943	7.6714	0.8634	10.6367	10.0338
		SVM	0.8914	16.1591	7.5079	0.8626	13.5980	11.4557
	Band + TF	MLR	0.7902	13.0310	10.3649	0.7455	12.5256	10.0958
		PLSR	0.0097	27.9304	18.4565	0.0812	18.2098	16.7097
		RF	0.7948	13.7665	9.0117	0.3395	15.1378	13.0904
		SVM	0.8889	16.9962	8.1721	0.4594	14.8344	12.3543
	Band + SF	MLR	0.5734	18.3716	14.2422	0.4043	14.4510	11.7219
		PLSR	0.5278	19.2859	14.9689	0.1590	17.1784	16.5621
		RF	0.7927	13.4065	8.1237	0.4619	16.0131	13.5150
		SVM	0.8948	15.8778	7.7449	0.5835	14.0798	12.6996
	Band + EV	MLR	0.8700	10.2435	8.4287	0.9286	8.2757	7.4666
		PLSR	0.4059	21.6330	15.0098	0.6302	13.3522	10.0488
		RF	0.8313	13.1994	7.6371	0.9523	9.6903	8.9120
		SVM	0.9478	14.9814	6.9915	0.7760	15.3824	12.8896

Table 3. Statistics on the results of the training and validation models. Band represents the model without any environmental variables, and EV represents the model with four environmental variables added at the same time.

worth mentioning that adding only the TF in the model constructed based on PLSR led to a significant decrease of 0.6123 (OSR), 0.3174 (FD), and 0.3364 (SD) in the accuracy of model estimation in the context of derivative processing. However, this phenomenon was not present in the other modeling methods. Furthermore, adding environmental variables, especially VF and CF, significantly improves the accuracy of the model constructed from the validation dataset, compared to the model without these variables. However, including only TF in “OSR + PLSR” lowers its accuracy ($R^2 = 0.0812$), which is notably inferior to that of the other models.

Although these environmental variables impacted the estimation model differently, adding them at the same time can significantly improve its fitting accuracy. Taking the model constructed based on the training dataset as an example, the accuracy of “OSR + EV + SVM” was 0.1702 higher than that of “OSR + Band + SVM”, 0.1756 higher than that of “OSR + VF + SVM”, 0.0203 higher than that of “OSR + CF + SVM”, 0.0250 higher than that of “OSR + TF + SVM”, and 0.0198 higher than that of “OSR + SF + SVM”. In summary, the combination of VF, CF, TF, and SF can significantly enhance SOC estimation accuracy within the confines of the study area. This was especially evident when the study covered a vast region and displayed substantial spatial variations in diverse environmental variables.

Modeling methods

Comparing the four modeling methods, it can be found that the conclusions of the models constructed based on the training and validation datasets were consistent. The results showed that under different derivative processing, both RF and SVM methods have higher accuracy in fitting and estimating SOC content compared with traditional regression analysis methods (MLR and PLSR). Specifically, the model based on the SVM method, with the original spectral bands and the four types of environmental variables achieved the highest accuracy (Training dataset: $R^2 = 0.9590$, RMSE = 13.9887, MAE = 10.8075; Validation dataset: $R^2 = 0.9220$, RMSE = 11.6165, MAE = 10.8075), followed by the “OSR + EV + RF” model. Specifically, considering the model developed based on the training dataset under identical derivative processing conditions, the accuracy of the “OSR + EV + SVM” model was 0.0856 greater than that of the “OSR + EV + RF” model. It exceeds the “OSR + EV + MLR” model by 0.1050 (RMSE = 11.0773, MAE = 9.2049) and the “OSR + EV + PLSR” model (RMSE = 21.6330, MAE = 15.0098) by 0.5531. Therefore, including environmental variables can effectively improve SOC accuracy, and the SVM method presents the best estimation accuracy, while the PLSR method is unsuitable for the study area. Besides, this study plotted the plots of the estimated and measured values of the sample points based on the SVM method (Fig. 5). The above results showed that including environmental variables significantly improved the estimation of SOC content, and the predicted values were in good agreement with measured values. Moreover, the

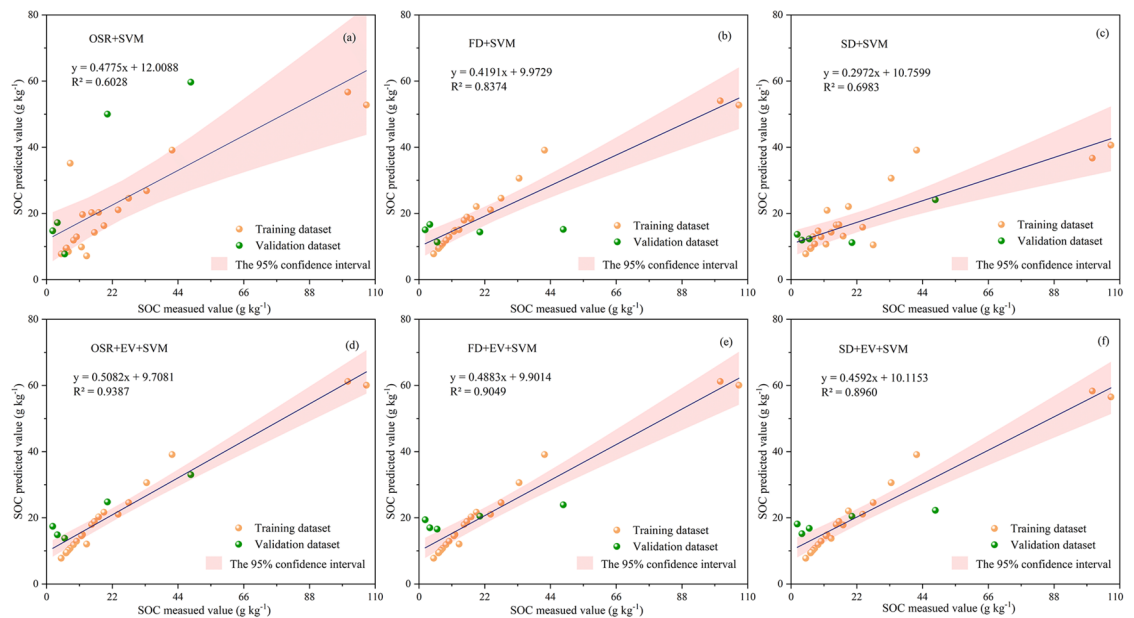


Figure 5. Comparison of measured and predicted SOC content among different SVM models.

“OSR + EV + SVM” model ($R^2 = 0.9387$) showed the best fit between the predicted and measured values (Fig. 5d). These results confirm that the “OSR + EV + SVM” model was the optimal SOC estimation model for the study area.

In general, environmental variables significantly increased the accuracy of the SOC content estimation model, and the best model was the “OSR + SVM” model combined with environmental variables, which had the best estimation performance. Derivative processing did not yield better estimation results. Among them, the effects of FD and OSR processing were similar, while the prediction effect of SD was weaker and significantly worse than that of the case without derivative processing. As for the choice of modeling methods, machine learning prediction has a higher accuracy than traditional methods. Among them, the SVM model has the most prediction ability, followed by RF, while the PLSR model is not suitable for fitting and predicting.

Best estimate modeling and spatial mapping of SOC content in the study area

Since the “OSR + EV + SVM” model had a higher prediction accuracy than other models, we used this model to estimate the SOC content in the study area (Fig. 6). Spatially, the SOC content in the study area in 2001 was highly spatially variable, with a mean value of 31.74 g kg^{-1} and a standard deviation of 28.31 g kg^{-1} . The central and western parts of the NECT had low SOC content while the eastern part had high SOC content. This trend was consistent with environmental variables, particularly elevation, slope, and precipitation. Statistical analysis of the SOC content indicated high SOC content in the entire study area. About 57.50% of the area had a SOC content higher than 20.00 g kg^{-1} . 23.7% of the area had SOC content higher than 50.00 g kg^{-1} , and 33.80% of the covered area had a SOC content between 20.00 and 50.00 g kg^{-1} . In addition, areas with SOC content above 100.00 g kg^{-1} and below 5.00 g kg^{-1} accounted for 2.30% and 19.90% of the total area, respectively.

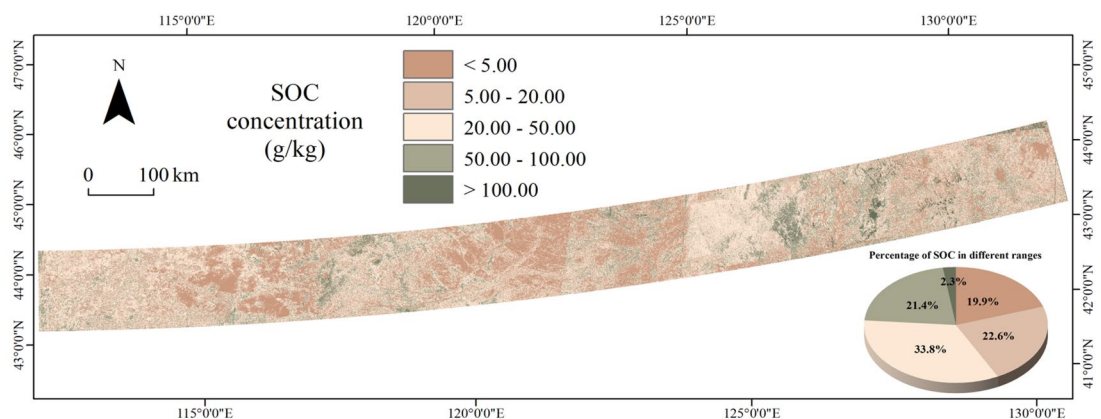


Figure 6. The SOC content in the study area based on the best model, 2001.

Discussion

Accurate quantification of soil organic carbon content is crucial for soil safety assessment, carbon cycling, and climate change research. Currently, constructing models to estimate SOC content based on optical remote sensing has become a pivotal research approach. However, the choice of different image processing methods, modeling techniques, and selection of environmental variables significantly influences model accuracy. Derivative processing aids in smoothing background information and reducing noise interference^{27,29,30}, widely applied in hyperspectral studies. We hypothesized that integrating Landsat 5 with environmental factors can effectively estimate SOC content, while derivative processing suppresses information expression in Landsat 5 images. Model comparison results validate our hypothesis that Landsat 5 images effectively convey SOC information, with derivative processing showing no significant improvement in SOC content estimation accuracy, consistent with previous studies^{13,60}. This may be attributed to the large coverage area, high variability in soil physicochemical properties, significant climate differences, and complex terrain and vegetation cover variations in the study area. Regarding modeling methods, studies by Bao et al.⁶¹ and Sabetizade et al.⁶² demonstrate that machine learning methods better handle the non-linear relationship between SOC content and spectral features, enhancing prediction reliability and predictability. Our findings support these conclusions. Additionally, our results indicate that Random Forest (RF) and Support Vector Machine (SVM) exhibit significant modeling advantages in estimating SOC content based on small sample datasets, particularly with SVM demonstrating stable modeling capabilities.

Our study reveals that adding environmental factors can effectively enhance SOC content estimation accuracy. Among these factors, mean annual precipitation and soil pH are the primary determinants influencing SOC content estimation accuracy, followed by elevation, slope, and aspect. This aligns with existing research^{4,28,31}. Temperature influences SOC content significantly through its effects on soil microbial activity and soil respiration^{63,64}. Soil pH affects soil chemical reactions and ion exchange properties, with acidic soils typically limiting microbial activity and organic matter decomposition, thereby inhibiting SOC accumulation⁶⁵. Precipitation indirectly affects SOC content by influencing soil moisture status. These findings confirm the significant impact of environmental factors on the spatial distribution of SOC content, with varied effects observed among different environmental variables.

In our study, integrating soil organic carbon data with Landsat 5 elucidates the influence of environmental variables on SOC content estimation. While our conclusions are insightful, they come with limitations. Firstly, the factors influencing spatial distribution of soil organic carbon content are complex, and this study only discusses common environmental variables. Future research will incorporate additional factors influencing spatial distribution of soil organic carbon content and analyze the differential importance of various environmental variables. For instance, soil clay particles adsorb and protect soil organic matter, preventing microbial decomposition or hydraulic erosion, often correlating with higher soil organic carbon content, a factor worth considering in future studies. Secondly, anthropogenic management practices also significantly influence soil organic carbon content. Subsequent studies should adequately characterize factors such as land use types and management practices (e.g., land use intensity) to enhance SOC content estimation capability. Lastly, our study covers a broad area with diverse land cover types, necessitating the collection of more sample points and the use of machine learning algorithms and feature selection suitable for big data to further analyze the impact of different environmental factors on soil organic carbon content across different regions, thereby improving SOC mapping. In the next step, we will integrate the latest satellite data with the results of a new round of large-scale sampling. This approach will enable us to include additional environmental variables and spectral bands for SOC mapping, thereby refining our methods and enhancing the accuracy of SOC content detection.

Conclusion

Through comparative analysis of derivative processing and four different modeling methods, our study generated soil organic carbon content maps for the surface layer of the Northeast Transect at a spatial resolution of 30 m. Results demonstrate that adding environmental variables enhances the accuracy of SOC content estimation across various models. Concerning derivative processing, the performance of OR and FD modeling methods showed minimal difference, while SD modeling exhibited poorer capabilities. In terms of modeling method selection, SVM models constructed with the inclusion of four categories of environmental factors consistently displayed strong predictive abilities, whereas Partial Least Squares Regression (PLSR) performed less effectively. Therefore, “OR + environmental variables + SVM” represents the optimal model for estimating SOC content in the study area ($R^2 = 0.9590$, RMSE = 13.9887, MAE = 10.8075). Our research highlights the importance of key environmental variables in monitoring SOC content. This study provides theoretical reference for the rapid estimation of large-area soil organic carbon content using remote sensing images combined with environmental factors. It also offers technical support for more precise SOC content detection in the future, which is crucial for soil conservation and sustainable development in Northeast China.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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X.X.: Formal analysis, Investigation, Methodology, Writing-original draft, Visualization. Q.H.: Conceptualization, Resources, Writing-review and editing, Validation, Supervision, Project administration, Funding acquisition. S.M.: Data collection, Data curation, Validation. J.L.: Software, Writing-review and editing. W.S.: Software. Y.L.: Data collection, Data curation. R.Y.: Data collection, Data curation. All authors have read and agreed to the published version of the manuscript. All the authors approved the final article.

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Competing interests

The authors declare no competing interests.

Additional information

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